

```

test> use FacultySystemDB
switched to db FacultySystemDB
FacultySystemDB> db.createCollection("students")
{ ok: 1 }
FacultySystemDB> db.students.insertOne({FirstName:"Ahmed",LastName:"Ali",Age:20,Faculty:{Name:"Computer Science",Address:"Cairo",Pass:true}},IsFired:false})
{
  acknowledged: true,
  insertedId: ObjectId('6a9dbd2ac53bbb10fb6d8f0c')
}
FacultySystemDB> db.students.insertMany([
  {FirstName:"Mohamed",LastName:"Hassan",Age:22,Faculty:{Name:"Engineering",Address:"Giza",Pass:true}},
  {FirstName:"Ahmed",Age:19,Faculty:{Name:"Medicine",Address:"Cairo"},Grades:[{CourseName:"Biology",Grade:55,Pass:true}],IsFired:false})
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a9dbd44c53bbb10fb6d8f0d'),
    '1': ObjectId('6a9dbd44c53bbb10fb6d8f0e')
  }
}
FacultySystemDB> db.students.find()
[
  {
    _id: ObjectId('6a9dbd2ac53bbb10fb6d8f0c'),
    FirstName: 'Ahmed',
    LastName: 'Ali',
    Age: 20,
    Faculty: { Name: 'Computer Science', Address: 'Cairo' },
    Grades: [
      { CourseName: 'Database', Grade: 90, Pass: true },
      { CourseName: 'OOP', Grade: 85, Pass: true }
    ],
    IsFired: false
  },
  {
    _id: ObjectId('6a9dbd44c53bbb10fb6d8f0d'),
    FirstName: 'Mohamed',
    LastName: 'Hassan',
    Age: 22,
    Faculty: { Name: 'Engineering', Address: 'Giza' },
    Grades: [ { CourseName: 'Math', Grade: 70, Pass: true } ],
    IsFired: false
  },
  {
    _id: ObjectId('6a9dbd44c53bbb10fb6d8f0e'),
    FirstName: 'Omar',
    LastName: 'Ahmed',
    Age: 19,
  }
]

```

```

]
FacultySystemDB> db.students.find({
|   FirstName: "Ahmed"})
|
|
| {
|   _id: ObjectId('6a9dbd2ac53bbb10fb6d8f0c'),
|   FirstName: 'Ahmed',
|   LastName: 'Ali',
|   Age: 20,
|   Faculty: { Name: 'Computer Science', Address: 'Cairo' },
|   Grades: [
|     { CourseName: 'Database', Grade: 90, Pass: true },
|     { CourseName: 'OOP', Grade: 85, Pass: true }
|   ],
|   IsFired: false
| }
|
]
FacultySystemDB> db.students.find({
|   $or: [
|     { FirstName: "Ahmed" },
|     { LastName: "Ahmed" }
|   ]
| })
|
|
| {
|   _id: ObjectId('6a9dbd2ac53bbb10fb6d8f0c'),
|   FirstName: 'Ahmed',
|   LastName: 'Ali',
|   Age: 20,
|   Faculty: { Name: 'Computer Science', Address: 'Cairo' },
|   Grades: [
|     { CourseName: 'Database', Grade: 90, Pass: true },
|     { CourseName: 'OOP', Grade: 85, Pass: true }
|   ],
|   IsFired: false
| },
| {
|   _id: ObjectId('6a9dbd44c53bbb10fb6d8f0e'),
|   FirstName: 'Omar',
|   LastName: 'Ahmed',
|   Age: 19,
|   Faculty: { Name: 'Medicine', Address: 'Cairo' },
|   Grades: [ { CourseName: 'Biology', Grade: 55, Pass: true } ],
|   IsFired: true
| }
|
]
FacultySystemDB> 

```

```

FacultySystemDB> db.students.find({
|   Age: { $lt: 21 }
| })
[
| {
|   _id: ObjectId('6a9dbd2ac53bbb10fb6d8f0c'),
|   FirstName: 'Ahmed',
|   LastName: 'Ali',
|   Age: 20,
|   Faculty: { Name: 'Computer Science', Address: 'Cairo' },
|   Grades: [
|     { CourseName: 'Database', Grade: 90, Pass: true },
|     { CourseName: 'OOP', Grade: 85, Pass: true }
|   ],
|   IsFired: false
| },
| {
|   _id: ObjectId('6a9dbd44c53bbb10fb6d8f0e'),
|   FirstName: 'Omar',
|   LastName: 'Ahmed',
|   Age: 19,
|   Faculty: { Name: 'Medicine', Address: 'Cairo' },
|   Grades: [ { CourseName: 'Biology', Grade: 55, Pass: true } ],
|   IsFired: true
| }
| ]
FacultySystemDB> db.students.find({
|   IsFired: true
| })
[
| {
|   _id: ObjectId('6a9dbd44c53bbb10fb6d8f0e'),
|   FirstName: 'Omar',
|   LastName: 'Ahmed',
|   Age: 19,
|   Faculty: { Name: 'Medicine', Address: 'Cairo' },
|   Grades: [ { CourseName: 'Biology', Grade: 55, Pass: true } ],
|   IsFired: true
| }
| ]
FacultySystemDB> db.students.find({
|   Age: { $gte: 21 },
|   Faculty: { $ne: null }
| })
[
| {
|   _id: ObjectId('6a9dbd44c53bbb10fb6d8f0d'),
|   FirstName: 'Mohamed',
|   LastName: 'Hassan',

```

```
]
FacultySystemDB> db.students.find(
|   { FirstName: "Ahmed" },
|   {
|       _id: 0,
|       FirstName: 1,
|       LastName: 1,
|       IsFired: 1
|   }
| )
[ { FirstName: 'Ahmed', LastName: 'Ali', IsFired: false } ]
FacultySystemDB> █
```

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]
[ { FirstName: 'Ahmed', LastName: 'Ali', IsFired: false } ]
```

```
FacultySystemDB> db.students.updateOne(
```

```
  { FirstName: "Ahmed" },
  {
    $set: {
      LastName: "Mostafa"
    }
  }
)
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
FacultySystemDB> db.students.createIndex({
```

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  FirstName: 1
})
```

```
FirstName_1
```

```
FacultySystemDB> db.students.getIndexes()
```

```
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { FirstName: 1 }, name: 'FirstName_1' }
]
```

```
FacultySystemDB> db.students.updateMany(
```

```
  {},
  {
    $unset: {
      "Faculty.Address": ""
    }
  }
)
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
FacultySystemDB>
```

```
]
FacultySystemDB> db.students.updateMany(
|   {},
|   {
|       $unset: {
|           "Faculty.Address": ""
|       }
|   }
| )
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 3,
|   modifiedCount: 3,
|   upsertedCount: 0
| }
FacultySystemDB> db.students.deleteMany({
|   IsFired: true
| })
| { acknowledged: true, deletedCount: 1 }
FacultySystemDB> db.students.drop()
true
FacultySystemDB> db.dropDatabase()
| { ok: 1, dropped: 'FacultySystemDB' }
FacultySystemDB>
```